Amit Chandran

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Mechanical Engineer with skills and experience in development of software, hardware and mechanical components within cross-functional teams. Aspiring to work within robotics, control systems or simulation.

Education

UCL - MEng Mechanical Engineering *Expected June 2028*

- Relevant Modules: Thermodynamics, Dynamics, Mathematical Modelling, Design, Materials
- Grade: Y1 – 85%

Experience

Systems Engineering Intern – Leonardo UK *Summer 2025 and 2026*

- Operated as a systems engineer, coordinating 10 other interns to seamlessly integrate antenna, firmware, hardware and software components into a continous RF to web-server pipeline.
- Implemented HDL-compatible signal processing pipelines in Simulink and deployed them on an RFSoC FPGA board, producing high-resolution spectra
- Developed a compression algorithm reducing FFT data bandwidth requirements by over **2500×**, minimising network utilisation whilst preserving critical information
- Architected a self-test algorithm to verify RF signal integrity, cutting downtime and boosting diagnostics
- Optimised signal max-hold algorithm using MATLAB to decrease run time by **99.3%**, reducing system latency by an order of magnitudes

Projects / Extracurricular

Technical Lead and Drone Controls Engineer – UCL Rover Team *January 2025 – present*

- Programmed an edge-computing vision pipeline in Python and OpenCV on a Raspberry Pi to track ArUco markers in real-time
- Established a robust UDP telemetry link to stream positional data seamlessly from the edge device to a high-level MATLAB/Simulink control model
- Interfaced the Simulink architecture directly with an Aero Selfie H743 flight controller to achieve an autonomous, end-to-end vision-to-flight pipeline

Smart Pet Feeder – BLE IoT System *October - March 2026*

- Developed bespoke C++ firmware for an ESP32 microcontroller utilising the NimBLE stack, achieving a **50%** reduction in flash usage versus the standard Bluedroid library
- Engineered an automatic dispensing mechanism, integrated with load-cells as well as RTC synchronization, achieving **±1%** accuracy for times and weights of dispensed food
- Implemented an event-driven GATT server architecture using the Nordic UART Service (NUS) enabling low latency communication with a custom Android client to load pet data

IMechE Design Challenge – Autonomous Robotic Device *January – March 2025*

- Led the team to develop a bespoke electro-mechanical solution involving H-bridges and latch circuits, maximising autonomy over competitors and placing 3rd of 45 teams
- Designed and assembled a PCB using EasyEDA, cutting weight by **85%** as well as reducing internal wire length by **1.2** metres versus breadboard designs
- Formulated a rigorous test strategy and built a test rig to rapidly implement upgrades to the vehicle
- Collaborated with mechanical designers to prioritise DFMA (Design for Manufacturing and Assembly), maximising space efficiency and vehicle modularity

Skills

- **Programming Languages** – MATLAB, Arduino, Python, C++
- **Tools / Software** – Simulink, Fusion 360, Git, Azure DevOps, EasyEDA
- **Manufacturing** – 3D Printing, Laser Cutting, Standard workshop equipment, Soldering